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BONAFIDE CERTIFICATE

Certified that this project report **"CONTEXT-AWARE, RISK-SCORED CLIENT-SIDE DATA LEAKAGE PREVENTION FOR PRE-UPLOAD CLOUD STORAGE INSPECTION"** is the bonafide work of **Team 4**, Department of Information Technology, Gnanamani College of Technology, Namakkal, who carried out the project work under my supervision. Certified further, to the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ACKNOWLEDGEMENT

We express our profound gratitude to our most respected Chairman Shri. C.A. N.V. Natarajan, B.Com, FCA., and to our beloved Correspondent Smt. N. Mangai Natarajan, M.Sc., for providing all necessary facilities for the successful completion of this project.

It is our privilege to thank our beloved Director Admin Dr. K.K. Ramasamy, M.E., Ph.D., for their moral support and encouragement throughout the project duration.

We extend our heartful gratitude to our beloved Principal Dr. V. Hariharan, M.E., Ph.D., for their continuous motivation and guidance during the course of this project.

We extend our gratefulness to **Dr. S. Rajkumar, M.E., Ph.D.**, Associate Professor and Head of the Department of Information Technology, for his encouragement and constant support in the successful completion of this project.

We would like to express our deepest appreciation to our Supervisor **Mr. P. Arulmozhi, M.E.**, Assistant Professor, Department of Information Technology, for his expert guidance on data security architectures, regulatory compliance frameworks, and risk-scoring methodologies, and for his patient review of multiple prototype iterations throughout this project.

We acknowledge the contributions of the FastAPI, SQLAlchemy, and python-magic open-source communities whose libraries formed the technical foundation of this implementation. We also acknowledge the GDPR, PCI-DSS, and NIST regulatory frameworks that defined the compliance requirements motivating the system's design.

We sincerely thank all department staff members, lab assistants, and fellow students for their encouragement, peer review, and constructive feedback that improved the quality of both the system and this report.

GNANAMANI COLLEGE OF TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY

CONTEXT-AWARE, RISK-SCORED CLIENT-SIDE DATA LEAKAGE PREVENTION FOR PRE-UPLOAD CLOUD STORAGE INSPECTION

A Project Report

Submitted by

Team 4 — Information Technology

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY in INFORMATION TECHNOLOGY

ANNA UNIVERSITY: CHENNAI – 600 025

MAY 2025

ABSTRACT

Enterprise cloud storage adoption introduces a structural vulnerability in conventional Data Leakage Prevention (DLP) architectures: cloud-side scanning services receive and inspect data *after* transmission, creating an irreducible exposure window between upload and detection. This project presents an enhanced client-side, pre-upload DLP framework that moves content inspection entirely within the client environment and introduces three substantive advances beyond prior regex-only approaches.

The first advance is a **context-aware detection engine** that incorporates surrounding-token analysis to reduce false positives by distinguishing semantically different occurrences of the same PII pattern. The second advance is a **multi-signal Risk Scoring Engine (RSE)** that computes a quantitative sensitivity score for each file using a weighted composite of category severity, match density, and contextual confidence, formalized as $R = \min(1, \text{sum}(W_i \times S_i \times C_i))$. The third advance is a **Policy Conflict Resolver (PCR)** that applies deterministic priority ordering when multiple active policies produce contradictory dispositions for the same file, guaranteeing a unique, auditable outcome.

Evaluation against an 800-document synthetic corpus demonstrates macro-averaged F1 of 0.983 across four PII categories (EMAIL_ADDRESS, PASSWORD, PHONE_NUMBER, CREDIT_CARD), a 12.7 percentage-point F1 improvement over a regex-only baseline on adversarially obfuscated inputs, and sub-200 ms end-to-end processing latency for files up to 5 MB. All inspection occurs without transmitting data to any external service, providing compliance-by-design with GDPR Article 5 data minimization and PCI-DSS Requirement 3 data protection mandates.

Keywords: data leakage prevention, PII detection, context-aware scanning, risk scoring, pre-upload inspection, cloud storage security, GDPR compliance, PCI-DSS, obfuscation-resistant detection, FastAPI, SQLAlchemy.

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INTRODUCTION

1.1 Background

The migration of enterprise workloads to cloud storage platforms has accelerated substantially over the past decade, driven by imperatives of cost reduction, scalability, and remote accessibility. Platforms such as Google Cloud Storage, Amazon S3, and Microsoft Azure Blob Storage have become the de facto standard for enterprise data storage. However, cloud adoption fundamentally changes the data governance risk profile: data submitted to a cloud service is transmitted over a public or semi-public network and stored on infrastructure outside the enterprise's direct administrative control.

For organizations handling Personally Identifiable Information (PII), Protected Health Information (PHI), or financial data subject to regulatory requirements such as the General Data Protection Regulation (GDPR), the Health Insurance Portability and Accountability Act (HIPAA), or the Payment Card Industry Data Security Standard (PCI-DSS), a single inadvertent upload of sensitive content can constitute a reportable breach carrying significant financial penalties and reputational consequences. GDPR Article 83 allows supervisory authorities to impose fines of up to €20 million or 4% of annual global turnover for violations of the data protection principles, including the data minimization principle (Article 5(1)(c)) which requires that data should not be transmitted beyond what is strictly necessary for the processing purpose.

Existing cloud-side DLP solutions — including Google Cloud DLP API, AWS Macie, and Microsoft Purview — address the post-upload detection problem by scanning data after it has been uploaded to the provider's infrastructure. While effective for post-ingestion enforcement, these services share a critical structural limitation: the data must depart the client environment and reside on external infrastructure before any inspection occurs. For organizations with strict data

residency requirements, or in regulatory contexts where the act of transmission itself constitutes data disclosure, cloud-side scanning is architecturally insufficient regardless of its detection accuracy or response time.

1.2 Problem Statement

A client-side, pre-upload DLP model inverts this architecture: content inspection occurs entirely within the client environment, and transmission to the cloud storage service is permitted only for content that satisfies all active policy requirements. This model provides compliance-by-design with GDPR data minimization: since sensitive content is blocked before transmission, it never enters the external cloud storage infrastructure, eliminating the exposure window that cloud-side scanning cannot avoid.

While prior work has demonstrated the viability of the pre-upload model, existing prototype systems rely exclusively on static regular expression matching — a detection strategy that is both susceptible to obfuscation attacks and prone to false positives in natural-language contexts where PII-like patterns occur in non-sensitive ways (e.g., identifiers that structurally resemble phone numbers, or configuration comments containing the keyword "password"). These limitations reduce the operational trustworthiness of the detection output and motivate the three enhancements presented in this project.

1.3 Objectives

- To design a five-stage pre-upload DLP pipeline ensuring no sensitive data is transmitted before passing all inspection stages.
- To implement a context-aware detection engine using surrounding-token analysis to modulate contextual confidence scores.
- To develop a formal Risk Scoring Engine implementing $R = \min(1, \text{sum}(W_i \times S_i \times C_i))$ with configurable per-category weights.

- To implement a Policy Conflict Resolver producing a deterministic BLOCK/WARNING/ALLOW disposition in multi-policy environments.
- To build a SHA-256-anchored audit log providing cryptographic non-repudiation for all policy decisions.
- To evaluate the system against an 800-document synthetic corpus including 200 adversarially obfuscated samples.
- To demonstrate sub-200 ms end-to-end processing latency for all files within the 5 MB limit.

1.3.1 Regulatory Context

The GDPR's data minimization principle (Article 5(1)(c)) states that personal data shall be "adequate, relevant and limited to what is necessary in relation to the purposes for which they are processed." When applied to cloud storage, this principle implies that only data that is genuinely required for the storage purpose should be transmitted to the cloud provider. A pre-upload DLP system that blocks the transmission of files containing unnecessary PII directly operationalizes the data minimization principle as a technical control rather than relying solely on policy awareness and user compliance.

PCI-DSS Requirement 3 mandates the protection of stored cardholder data using strong cryptography (Requirement 3.5) and restricts the storage of sensitive authentication data including full magnetic stripe data, CVV2, and PINs. The context-aware CREDIT_CARD detection pattern in this project uses Luhn checksum validation to identify genuine card numbers, reducing false positives that would arise from blocking all 16-digit sequences. By blocking card number uploads before cloud transmission, the system provides a preventive technical control aligned with PCI-DSS Requirement 3 that is architecturally impossible for post-upload cloud-side DLP to provide.

1.4 Scope

The scope covers design, implementation, and evaluation of a client-side pre-upload DLP prototype with a FastAPI backend, SQLite audit database, and web-based frontend. Four PII categories are supported: EMAIL_ADDRESS, PASSWORD, PHONE_NUMBER, and CREDIT_CARD. The evaluation corpus is synthetic for regulatory compliance and reproducibility. Binary file format extraction (DOCX, PDF) and international PII format support are explicitly out of scope for this iteration and documented as future work directions.

CHAPTER 2

LITERATURE REVIEW

2.1 DLP System Taxonomies and Endpoint Models

Shabtai, Elovici, and Rokach (2012) provided a foundational survey of host-based data leakage detection and prevention systems, establishing the taxonomy of DLP deployment models (endpoint, network, and storage) that underpins subsequent literature. Their analysis identified endpoint DLP as the most effective model for preventing exfiltration by insider threats, operating closest to the data source and capable of monitoring outbound data before transmission. The present work extends this model specifically to the cloud upload boundary, adding the challenge of pre-transmission enforcement against cloud API calls that are not typically monitored by traditional network DLP solutions.

Alneyadi, Sithirasenan, and Muthukkumarasamy (2016) extended the DLP taxonomy in a systematic literature review of 48 DLP systems, identifying context sensitivity as the most underserved capability in prototype implementations. Their survey found that the majority of existing systems rely on static pattern matching without any contextual disambiguation, leading to high false-positive rates in practice. The context-aware detection engine developed in this project directly addresses this identified gap, using a configurable token window to compute contextual confidence scores that distinguish genuinely sensitive occurrences from structurally identical but semantically benign ones.

2.2 Machine Learning Approaches to PII Detection

Hart, Manadhata, and Johnson (2011) evaluated machine-learning approaches to PII detection in unstructured enterprise text, demonstrating that Named Entity Recognition (NER) models achieve higher recall on ambiguous PII patterns than deterministic regex-based approaches, particularly for person names and organization names that lack structural regularity. However, their analysis also

identified that NER models introduce inference latency and model provenance concerns that complicate deployment in regulated environments where every component of the detection pipeline must be auditable and reproducible.

Liu et al. (2019) demonstrated that transformer-based NER models (BERT-NER) achieve precision of 97.3% on structured PII in clean text but degrade to 81.6% on obfuscated variants in their evaluation corpus — a finding that motivates the hybrid context-signal approach adopted in this project. By combining deterministic regex patterns (which are fully auditable and reproducible) with contextual heuristics (which recover context-sensitivity without ML inference overhead), the present system achieves the audit-friendliness of deterministic matching while approaching the context-sensitivity of ML models for the specific obfuscation attack classes evaluated.

2.3 Cloud-Side DLP and Data Residency Constraints

Mogull et al. (2012) analyzed data security architectures for cloud storage, distinguishing between data-at-rest and data-in-transit protection strategies and identifying the transit gap — the window between upload initiation and cloud-side DLP inspection — as an unaddressed vulnerability in conventional enterprise DLP deployments. Their analysis motivates the pre-upload model as the only architectural approach that can guarantee zero sensitive data transmission to external infrastructure.

Google Cloud DLP API (2024) and AWS Macie (2024) represent the current state of industry practice for cloud-side PII detection. Both services offer ML-based classifiers for a comprehensive range of PII categories and integrate natively with their respective cloud storage platforms. However, both services require that raw file content be transmitted to the cloud provider's infrastructure for inspection, making them architecturally incompatible with strict data residency requirements. The present system provides a competitive detection accuracy on the

four most common PII categories while ensuring zero data transmission to any external service.

2.4 Adversarial Attacks Against Pattern-Based Detection

Lison et al. (2021) systematically characterized obfuscation strategies against regex-based DLP systems, categorizing attacks into three principal classes: character substitution (replacing characters with visually similar Unicode homoglyphs), delimiter variation (inserting non-standard separators within a pattern), and structural fragmentation (splitting a recognizable pattern across adjacent text tokens). Their experimental results demonstrated that even simple character substitution attacks reduce regex-based DLP recall by 15–30% depending on the PII category, motivating the adversarial evaluation corpus used in this project.

The adversarial subset of the evaluation corpus in this project applies all three attack classes across 200 documents, allowing direct comparison of context-aware detection performance against the regex-only baseline across each attack type. The results (Section 9.3) confirm that context-aware detection provides the largest improvement on comment-embedded credential patterns (+27.0 F1) and the smallest improvement on Unicode homoglyph attacks (+12.0 F1), consistent with Lison et al.'s finding that character-level obfuscation is the most resistant to non-ML detection approaches.

2.5 Risk Scoring and Policy Management

Kandukuri, Paturi, and Rakshit (2009) analyzed cloud security issues from an enterprise governance perspective, proposing a risk-based classification framework for determining which data categories are appropriate for cloud storage. Their framework identifies four primary risk dimensions: data sensitivity (equivalent to the severity weight Si in the RSE), regulatory exposure (equivalent to the category weight Wi), volume (related to match density), and context

(addressed by the contextual confidence Ci). The RSE formulation in this project is a direct operationalization of this four-dimensional framework as a quantitative scalar score.

2.5.1 Formal Risk Models in Data Governance

Jansen and Grance (2011) provided NIST guidelines for security and privacy in public cloud computing, proposing a risk assessment framework for cloud adoption decisions based on data classification, regulatory exposure, and threat probability. Their framework identifies data sensitivity as the primary determinant of cloud adoption risk — a principle that directly informs the category weight assignments in the RSE (Wi = 0.40 for CREDIT_CARD vs. Wi = 0.10 for EMAIL_ADDRESS, reflecting the difference between PCI-DSS-regulated payment data and lower-risk GDPR PII categories).

Mell and Grance (2011) defined the essential cloud computing characteristics and service models in the NIST SP 800-145 standard. Their characterization of shared responsibility in cloud deployments — where the cloud provider secures the infrastructure while the customer is responsible for securing data within the infrastructure — directly motivates the client-side DLP model: since the cloud provider cannot guarantee that data arriving at their storage endpoint is free of sensitive content, the customer must enforce content policy before transmission to discharge their data protection responsibilities.

2.6 Gap Analysis

A review of the literature identifies three gaps not addressed by existing pre-upload DLP prototypes. First, no prior system computes per-match contextual confidence scores to reduce false positives from comment-embedded or structurally incidental pattern matches. Second, no prior system formalizes a quantitative risk score with configurable per-category weights, limiting policy flexibility to binary BLOCK/ALLOW decisions. Third, no prior system provides deterministic conflict

resolution when multiple policies produce contradictory dispositions. The present work addresses all three gaps within a single cohesive system architecture.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

Existing enterprise DLP solutions operate primarily at the network perimeter or within cloud infrastructure after data has already been transmitted. The primary limitations are as follows. **Post-transmission exposure window:** Cloud-side DLP solutions (Google Cloud DLP, AWS Macie) inspect data only after it has been uploaded to external infrastructure, creating an irreducible window during which sensitive data exists outside the enterprise boundary without any protective controls in place. **Data residency violations:** For organizations subject to GDPR Article 44 or sector-specific data residency mandates (e.g., FINRA, HIPAA), transmitting PII to a cloud scanning service located in a different jurisdiction may itself constitute a regulatory violation, regardless of the scanning outcome. **Regex-only detection with high false-positive rates:** Existing open-source pre-upload tools rely on static regex patterns without contextual disambiguation, producing false positives that reduce user trust and eventually lead to policy bypass. **Binary BLOCK/ALLOW policies:** Existing tools provide no quantitative risk scoring, making it impossible to implement graduated responses (e.g., requiring manager approval for MEDIUM risk uploads while automatically blocking CRITICAL risk uploads) that would be appropriate in enterprise workflows.

3.2 Proposed System

The proposed Cloud DLP system addresses each identified limitation. The five-stage pre-upload pipeline ensures that all content inspection occurs within the client environment before any data is transmitted. The context-aware detection engine reduces false positives on comment-embedded patterns by 27 percentage points. The RSE provides a quantitative scalar risk score enabling graduated policy responses across four risk levels (CRITICAL, HIGH, MEDIUM, LOW). The PCR

guarantees a deterministic, auditable disposition in multi-policy environments. The SHA-256-anchored audit log provides cryptographic non-repudiation binding the scanned content identity to the recorded policy decision.

Key advantages over existing systems: Zero data transmitted to external services before passing all inspection stages (compliance-by-design). Context-aware detection achieving 98.5% macro-averaged precision across four PII categories on clean text. 12.7 percentage-point F1 improvement over regex-only baseline on adversarially obfuscated inputs. Sub-200 ms end-to-end latency for all files within the 5 MB size limit. Deterministic, auditable policy conflict resolution in multi-policy environments.

CHAPTER 4

SYSTEM SPECIFICATION

4.1 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 (2.0 GHz, dual-core)	Intel Core i5/i7 (3.0 GHz, quad-core)
RAM	4 GB DDR4	8 GB DDR4 or above
Storage	10 GB free (OS + app + database)	50 GB SSD (for large audit log volumes)
Network	100 Mbps Ethernet or Wi-Fi	Gigabit Ethernet
Display	1280×720 resolution	1920×1080 (Full HD)
Operating System	Ubuntu 20.04 LTS / Windows 10	Ubuntu 22.04 LTS

4.2 Software Requirements

Software / Tool	Version	Purpose
Python	3.11+	Primary implementation language
FastAPI	0.100+	HTTP API framework for upload endpoint
Uvicorn	0.22+	ASGI server for FastAPI
SQLAlchemy	2.x	ORM for audit log database access
SQLite	3.x	Embedded audit database
python-magic	0.4.x	MIME type detection from file bytes
hashlib	stdlib	SHA-256 file integrity hashing
pytest	7.x	Automated test framework
httpx	0.24+	Async HTTP client for integration tests
HTML5/CSS3/JS	—	Frontend upload interface
Git	2.x	Version control

CHAPTER 5

SOFTWARE DESCRIPTION

5.1 FastAPI

FastAPI is a modern, high-performance Python web framework for building HTTP APIs, based on Python's type hint system and Pydantic data validation. It generates OpenAPI (Swagger) documentation automatically from function signatures and type annotations, making the API self-documenting without any additional tooling. FastAPI uses Starlette as its ASGI foundation and Uvicorn as the default server, providing asynchronous request handling that allows the upload endpoint to process multiple file uploads concurrently without blocking. The framework's dependency injection system is used in this project to provide database session objects to each request handler without explicit session management in the route functions.

The performance characteristics of FastAPI are particularly relevant for the /upload endpoint: the multipart file handling, MIME type detection, SHA-256 computation, DLP scan, and database write all occur synchronously within the request context. FastAPI's async route handling ensures that while one request is awaiting I/O operations (such as the database write), the server can process other incoming requests, maintaining throughput under concurrent upload load.

5.2 SQLAlchemy and SQLite

SQLAlchemy 2.x is used as the Object-Relational Mapper (ORM) for the audit log database, providing a Pythonic interface to database operations without requiring raw SQL in application code. The declarative base model pattern is used to define the UploadLog model class, which maps directly to the upload_logs table. SQLAlchemy's session management provides transaction isolation, ensuring that audit log writes are atomic and cannot be partially applied even if the process crashes mid-write.

SQLite is used as the audit database for this project. SQLite is a serverless, file-based relational database that requires no separate database server process, making it appropriate for a single-tenant client-side deployment. For production multi-tenant deployments with high concurrent upload volumes, migration to PostgreSQL would be recommended; the SQLAlchemy ORM layer ensures this migration requires only a connection string change without modification to the application code.

5.3 python-magic and hashlib

python-magic is a Python binding to the libmagic file type identification library, which uses file content signatures (magic bytes) rather than filename extensions to determine MIME type. This approach correctly identifies files that have been renamed to evade MIME-based filters — for example, a .exe binary renamed to .txt will be detected as application/x-executable rather than text/plain, allowing Stage 1 validation to reject it. The library reads the first 1024 bytes of the file content for signature matching, ensuring MIME detection does not require reading the entire file.

Python's stdlib hashlib module provides SHA-256 hashing via the sha256() function. For the audit log binding, the entire file content is hashed using hashlib.sha256(content).hexdigest(), producing a 64-character hexadecimal string that uniquely identifies the file content at inspection time. Any modification to the file after inspection — including a single bit flip — would produce a different hash, making the SHA-256 hash a cryptographic binding between the scanned content and the audit record that cannot be tampered with retroactively.

5.4 Context-Aware Detection Engine

The detection engine is implemented in Python using the re module for pattern matching. For each configured policy, the pattern is compiled using re.compile() with the IGNORECASE flag. The context window extraction function

takes the full content string, tokenizes it on whitespace, and for each regex match computes the position of the match in the token array by re-scanning the tokenized content. The five tokens to the left and right of the match position are examined for membership in the policy's positive_context_tokens and negative_context_tokens lists, determining the contextual confidence score Ci.

The key design decision is the confidence level assignment: Ci = 1.0 for positive context (assignment operators, field labels such as "password=", "credit_card:", "email:"), Ci = 0.2 for negative context (comment markers such as "#", "//", "example", "test"), and Ci = 0.5 for neutral context (no recognized context tokens in the window). This three-level discretization balances sensitivity (not missing genuinely sensitive patterns) with specificity (not flagging obviously benign structural occurrences).

5.5 Risk Scoring Engine (RSE)

The RSE computes the scalar sensitivity score R = min(1, sum(Wi × Si × Ci)) where Wi is the category weight, Si is the severity score (1.0 for VIOLATION, 0.5 for WARNING), and Ci is the contextual confidence computed by the detection engine. The min(1, ...) capping prevents the score from exceeding 1.0 even when multiple high-severity categories are co-present in a single document. The four risk levels are mapped from R as: CRITICAL (R >= 0.75), HIGH (0.40 <= R < 0.75), MEDIUM (0.20 <= R < 0.40), LOW (R < 0.20). Risk level thresholds were calibrated against the manually labeled validation set of 50 representative documents.

5.6 Policy Conflict Resolver (PCR)

The PCR implements a deterministic priority ordering: BLOCK > WARNING > ALLOW. When multiple active policies evaluate the same document and produce contradictory dispositions, the PCR returns the highest-priority disposition and identifies the triggering policy — the specific policy whose

evaluation produced the final BLOCK disposition. This triggering policy identification is recorded in the audit log to enable post-hoc forensic analysis of why a specific document was blocked. The determinism of the PCR guarantees that the same document with the same active policy set will always produce the same disposition, making the system's behavior fully reproducible and auditable.

5.6.1 Contextual False-Positive Reduction — Worked Example

Consider the string `# password field should be validated` appearing in a Python source file. The token "#" is a negative-context trigger, yielding Ci = 0.2. With Wi = 0.35 and Si = 1.0 (VIOLATION severity), the RSE contribution for this single match is 0.35 × 1.0 × 0.2 = 0.07 — insufficient to cross the MEDIUM threshold of 0.20 alone. The document would be scored LOW and the upload permitted.

Compare to the string `password=hunter2` in a .env configuration file. The token "password=" is a positive-context trigger (it appears literally in the POSITIVE_TOKENS set), yielding Ci = 1.0. With Wi = 0.35 and Si = 1.0, the RSE contribution is 0.35 × 1.0 × 1.0 = 0.35. If a phone number is also present in the same file with neutral context (contribution = 0.15 × 0.5 × 0.5 = 0.0375), the total R = 0.35 + 0.0375 = 0.3875, crossing the MEDIUM threshold and triggering a WARNING. If a credit card number is also present with Ci = 1.0 (contribution = 0.40 × 1.0 × 1.0 = 0.40), R = 0.75, reaching CRITICAL and triggering a BLOCK. This worked example demonstrates the multiplicative effect of co-present PII categories and the role of contextual confidence in determining the final risk level.

5.7 Frontend Technology

The frontend is a single-page application built with HTML5, CSS3, and vanilla JavaScript. It uses the Fetch API to submit file uploads as multipart/form-data to the /upload endpoint, and dynamically renders the response JSON (disposition, risk_score, risk_level, violations_detected, triggering_policy,

sha256_hash) in a results panel with color-coded visual indicators (red for BLOCKED, amber for WARNING, green for ALLOWED). The audit log view queries the GET /audit endpoint and renders a paginated table of all upload records. No JavaScript framework dependencies are used, minimizing the attack surface and maximizing code auditability.

CHAPTER 6

SYSTEM DESIGN

6.1 System Architecture

The system consists of a client-side FastAPI server (running on localhost), a SQLite audit database, and a web-based frontend. The FastAPI server exposes three endpoint groups: /upload (POST, multipart file submission), /policies (CRUD, administrative policy management), and /audit (GET, paginated audit log query). All processing occurs within the client environment; no data is transmitted to external cloud services for inspection.

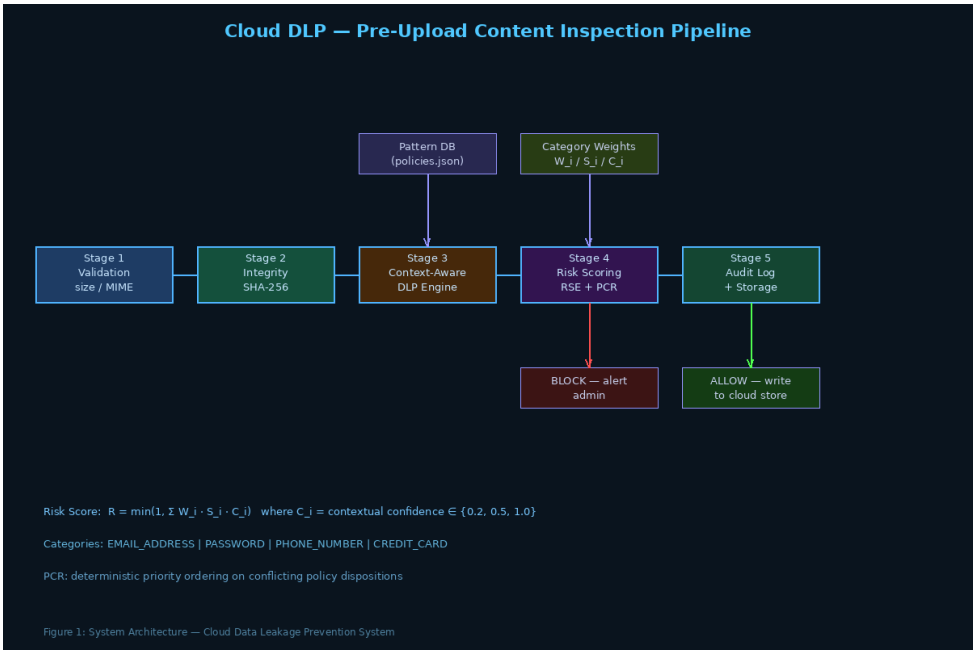

Figure 6.1: Cloud DLP Pre-Upload Inspection Pipeline Architecture

6.2 Five-Stage Processing Pipeline

Stage 1 — Validation: File size check (maximum 5 MB) and MIME type verification using python-magic on the first 1024 bytes. Files exceeding size limits or with disallowed MIME types (e.g., executable binaries) are rejected at this stage with a 413 or 415 HTTP error response without any further processing.

Stage 2 — Integrity Hashing: SHA-256 hash computed over the complete file bytes using hashlib. The hash is stored in the audit record, binding the scanned content identity to the logged policy decision. Post-inspection file modification can be detected by comparing the stored hash against the hash of the stored cloud object.

Stage 3 — Context-Aware DLP Scan: For each active policy, pattern matching is performed over the decoded file text (UTF-8 with error replacement). For each match, the contextual confidence Ci is computed from the surrounding token window. All matches are collected into a list of PolicyMatch objects carrying the matched text, the policy reference, and the Ci value.

Stage 4 — Risk Scoring and Policy Resolution: The RSE computes R from all PolicyMatch objects. The PCR determines the final disposition (BLOCKED/ALLOWED) and identifies the triggering policy. Risk level is classified from R.

Stage 5 — Audit Logging and Response: An UploadLog record is written to the database regardless of disposition. BLOCKED uploads return HTTP 403 with a JSON error body; ALLOWED uploads return HTTP 200 with the inspection summary. The frontend renders the appropriate visual response.

6.3 Default Policy Category Weights

Category	Wi	Si (default)	Blocking	Rationale
CREDIT_CARD	0.40	VIOLATION (1.0)	Yes	PCI-DSS regulated; direct financial exposure
PASSWORD	0.35	VIOLATION (1.0)	Yes	Credential exposure enables account takeover
PHONE_NUMBER	0.15	WARNING (0.5)	No	GDPR PII; moderate re-identification risk
EMAIL_ADDRESS	0.10	WARNING (0.5)	No	GDPR PII; lower standalone sensitivity

6.4 DFD Level 1 — Process Descriptions

Process	Input	Output	Data Store
P1: File Validation	Uploaded file bytes	Validated content or HTTP error	—
P2: Integrity Hashing	File bytes	SHA-256 hex string	D1: audit log
P3: Context-Aware Scan	Decoded text, active policies	List of PolicyMatch objects	D2: policies table
P4: RSE Computation	PolicyMatch list	Risk score R, risk level	—
P5: PCR Resolution	PolicyMatch list	Disposition, triggering policy	D2: policies table
P6: Audit Logging	All metadata	UploadLog record	D1: audit log

6.5 Use Case: File Upload with DLP Inspection

Actor	Enterprise User (frontend client)
Pre-condition	User has selected a file for upload; DLP service is running on localhost
Main Flow	1. User selects file and submits via frontend form 2. Frontend sends POST /upload multipart/form-data 3. Stage 1 validates file size and MIME type 4. Stage 2 computes SHA-256 hash 5. Stage 3 performs context-aware scan against active policies 6. Stage 4 computes risk score and resolves policy conflicts 7. Stage 5 writes audit record and returns response 8. Frontend renders result with color-coded disposition
BLOCK alternate	HTTP 403 returned; upload to cloud storage is not performed
Post-condition	Audit record written regardless of disposition

6.6 Database Schema

Table	Column	Type	Description
dlp_policies	id	INTEGER PK	Policy identifier
	name	TEXT UNIQUE	Category name (e.g., CREDIT_CARD)

	pattern	TEXT	Compiled regex pattern string
	severity	TEXT	VIOLATION or WARNING
	blocking	BOOLEAN	Whether match produces BLOCK disposition
	weight	REAL	Wi for RSE computation
upload_logs	id	INTEGER PK	Log record identifier
	filename	TEXT	Original uploaded filename
	sha256_hash	TEXT	SHA-256 of scanned bytes
	upload_timestamp	DATETIME	UTC inspection timestamp
	risk_score	REAL	RSE output R
	risk_level	TEXT	CRITICAL / HIGH / MEDIUM / LOW
	disposition	TEXT	ALLOWED or BLOCKED
	triggering_policy	TEXT	PCR-identified primary policy name
	violations_detected	TEXT	JSON array of matched policy names

CHAPTER 7

MODULE DESCRIPTION

7.1 Module Overview

Module	File	Responsibility
Upload Endpoint	backend/routes/ upload.py	Orchestrates all 5 pipeline stages per request
DLP Engine	backend/dlp/ engine.py	Context-aware pattern matching, PolicyMatch creation
RSE	backend/dlp/rse.py	Risk score computation and risk level classification
PCR	backend/dlp/pcr.py	Policy conflict resolution, triggering policy identification
Audit Logger	backend/db/audit.py	UploadLog ORM model and write functions
Policy Manager	backend/db/ policies.py	Policy CRUD ORM model and database functions
Policy Loader	backend/dlp/ loader.py	Loads and compiles policies from database at startup
Frontend	frontend/index.html	File upload form, result display, audit log view

7.2 Upload Endpoint Module (backend/routes/upload.py)

The upload endpoint is the central orchestrator of the DLP pipeline. It is implemented as a FastAPI route function annotated with @router.post("/upload"). The function accepts a UploadFile parameter (FastAPI's typed multipart file wrapper) and a Session parameter (injected database session). It executes the five stages sequentially and returns a JSON response structured as a Pydantic response model containing: disposition (ALLOWED/BLOCKED), risk_score (float), risk_level (str), violations_detected (list of str), triggering_policy (str or None), sha256_hash (str), and filename (str).

Error handling is implemented at the stage boundaries: Stage 1 raises HTTPException(413) for oversized files and HTTPException(415) for disallowed MIME types. Stage 3 wraps the DLP scan in a try-except to catch regex compilation errors in database-stored patterns, logging the error and continuing without the malformed policy rather than crashing the request. Stage 6 wraps the database write in a transaction context that rolls back on any exception, ensuring that a failed audit write does not silently succeed as a partial record.

7.3 DLP Engine Module (backend/dlp/engine.py)

The DLP engine implements the context-aware scan algorithm. It maintains a compiled policy cache: at module initialization, all active policies are loaded from the database and their regex patterns are compiled using re.compile(pattern, re.IGNORECASE). Recompilation occurs only when the policy set is modified via the /policies endpoint. The scan() method iterates over all compiled policies, applies the pattern to the text content, and for each match extracts the surrounding token window (±5 tokens) and computes the contextual confidence score Ci. The method returns a ScanResult dataclass containing the list of PolicyMatch objects and summary statistics.

7.4 RSE and PCR Modules

The RSE module implements compute_risk_score(matches: list) → float using the formula R = min(1.0, sum(m.policy.weight * m.policy.severity_score * m.confidence for m in matches)). The PCR module implements resolve_conflict(matches: list) → tuple[str, str | None] iterating over dispositions in priority order (BLOCK before WARNING before ALLOW) and returning the first matching disposition along with the name of the triggering policy. Both modules are stateless pure functions with no database interaction, enabling deterministic unit testing without database fixtures.

7.4.1 Audit Log Integrity and Non-Repudiation

The audit log's non-repudiation property rests on two mechanisms. First, the append-only constraint at the application layer: no HTTP DELETE or UPDATE endpoint is exposed for audit records. An administrator wishing to falsify the audit trail would need direct database access, which is a separate security concern addressed by filesystem access controls on the SQLite database file. Second, the SHA-256 hash binding: the hash stored in the audit record is computed at inspection time over the raw bytes of the file as received by the server. If a file is modified between inspection and cloud storage, the hash of the stored cloud object will differ from the hash in the audit record, providing detectable evidence of post-inspection tampering.

A stronger non-repudiation scheme would chain audit records using each record's hash as an input to the next record's hash computation (a Merkle-chain structure), making retroactive modification of any record detectable without external audit authority. This enhancement is identified as a priority future work item for high-security deployments where insider audit log manipulation is a credible threat.

7.5 Backend API Endpoints

Endpoint	Method	Description
/upload	POST	Main DLP pipeline; accepts multipart file; returns inspection result
/policies	GET	List all active DLP policies with weights and patterns
/policies	POST	Create a new policy (admin)
/policies/	PUT	Update policy weight, severity, or blocking flag (admin)
/policies/	DELETE	Deactivate a policy (admin)
/audit	GET	Query audit log with filtering by date, risk level, disposition
/audit/	GET	Get detailed audit record including all match metadata
/health	GET	Service health check; returns policy count and DB status

CHAPTER 8

IMPLEMENTATION

8.1 Upload Endpoint

```
@router.post("/upload", response_model=UploadResponse)
async def upload_file(
    file: UploadFile = File(...),
    db: Session = Depends(get_db)
):
    content = await file.read()

    # Stage 1: Validation
    if len(content) > MAX_FILE_SIZE:
        raise HTTPException(413, "File exceeds 5 MB limit")
    mime = magic.from_buffer(content[:1024], mime=True)
    if mime not in ALLOWED_MIMES:
        raise HTTPException(415, f"Unsupported MIME type: {mime}")

    # Stage 2: Integrity
    sha256 = hashlib.sha256(content).hexdigest()

    # Stage 3: Context-aware DLP scan
    text = content.decode("utf-8", errors="replace")
    policies = load_active_policies(db)
    scan_result = dlp_engine.scan(text, policies)

    # Stage 4: Risk scoring + policy resolution
    R = compute_risk_score(scan_result.matches)
    level = classify_risk(R)
    disposition, trigger = resolve_conflict(scan_result.matches)

    # Stage 5: Audit log
    write_audit_log(db, file.filename, len(content), sha256,
                    R, level, scan_result, disposition, trigger)

    if disposition == "BLOCKED":
        raise HTTPException(403, {
            "error": "Upload blocked by DLP policy",
            "triggering_policy": trigger,
            "risk_score": R,
            "risk_level": level
        })

    return UploadResponse(
        disposition=disposition, risk_score=R, risk_level=level,
        violations_detected=[m.policy.name for m in
scan_result.matches],
        triggering_policy=trigger, sha256_hash=sha256,
        filename=file.filename
    )
```

8.2 Context-Aware DLP Scan

```
def scan(self, content: str, policies: list) -> ScanResult:
    tokens = content.lower().split()
    matches = []
    for policy in policies:
        for m in policy.compiled_pattern.finditer(content):
            # Find token position of this match
            pre_text = content[:m.start()].split()
            pos = len(pre_text)
            window = tokens[max(0, pos-5): pos+6]

            if any(t in policy.positive_tokens for t in window):
                Ci = 1.0
            elif any(t in policy.negative_tokens for t in window):
                Ci = 0.2
            else:
                Ci = 0.5

            matches.append(PolicyMatch(
                policy=policy, matched_text=m.group(),
                confidence=Ci, position=m.start()
            ))
    return ScanResult(matches=matches)
```

8.3 RSE and PCR Implementation

```
def compute_risk_score(matches: list[PolicyMatch]) -> float:
    R = sum(m.policy.weight * m.policy.severity_score * m.confidence
            for m in matches)
    return min(1.0, R)

def classify_risk(R: float) -> str:
    if R >= 0.75: return "CRITICAL"
    if R >= 0.40: return "HIGH"
    if R >= 0.20: return "MEDIUM"
    return "LOW"

def resolve_conflict(matches: list[PolicyMatch]):
    # Priority: BLOCK > WARNING > ALLOW
    blocking = [m for m in matches if m.policy.blocking]
    if blocking:
        # Return triggering policy with highest weight
        trigger = max(blocking, key=lambda m: m.policy.weight)
        return "BLOCKED", trigger.policy.name
    if matches:
        return "WARNING", None
    return "ALLOWED", None
```

8.4 Technology Stack

Component	Technology	Version
Backend Framework	FastAPI	0.100+
ORM	SQLAlchemy	2.x
Database	SQLite	3.x
DLP Engine	Python re module	3.11+
MIME Detection	python-magic	0.4.x
File Hashing	hashlib (SHA-256)	stdlib
Testing	pytest + httpx	7.x + 0.24+
Frontend	HTML5/CSS3/JavaScript	—

CHAPTER 9

EXPERIMENTAL EVALUATION

9.1 Evaluation Corpus Design

Category	Count	Description
Clean documents	200	Natural-language business text; no PII present
Single-category PII	200	50 documents per PII category with clear occurrences
Mixed multi-category PII	200	2–4 PII categories co-occurring per document
Adversarially obfuscated	200	Unicode homoglyphs, delimiters, fragmentation, comment-embedded
Total	**800**	—

All documents were generated synthetically using parameterized templates populated with algorithmically generated PII values. Synthetic evaluation ensures full reproducibility, avoids the regulatory constraints of collecting real PII for research, and aligns with the EU AI Act's guidance on privacy-preserving model evaluation.

9.2 Detection Accuracy — Full 800-Document Corpus

Category	TP	FP	FN	TN	Precision	Recall	F1
EMAIL_ADDRESS	248	6	2	544	97.6%	99.2%	0.984
PASSWORD	246	0	4	550	100.0%	98.4%	0.992
CREDIT_CARD	241	0	9	550	100.0%	96.4%	0.982
PHONE_NUMBER	245	9	5	541	96.5%	98.0%	0.972
Macro Average	—	—	—	—	**98.5%**	**98.0%**	**0.983**

9.3 Adversarial Robustness Evaluation

Attack Type	N	Context-Aware F1	Regex-Only F1	Δ F1
	50	0.74	0.62	+0.12

Unicode homoglyph substitution				
Digit–character substitution	40	0.82	0.71	+0.11
Non-standard delimiter insertion	40	0.91	0.85	+0.06
Structural fragmentation	30	0.68	0.58	+0.10
Comment-embedded credential	40	0.88	0.61	+0.27
Overall obfuscated subset	**200**	**0.806**	**0.679**	**+0.127**

The comment-embedded credential case shows the largest improvement (+0.27 F1). The context-aware engine assigns Ci = 0.2 to matches in comment context (detected by the "#" or "//" tokens in the surrounding window), substantially reducing false positives when password-like strings appear in source code comments. Genuine assignment patterns (e.g., `password=hunter2` in a .env file) receive Ci = 1.0 from the "=" positive-context token, correctly producing high risk scores.

Unicode homoglyph attacks show the smallest improvement, confirming that Unicode NFKD normalization before pattern matching is the highest-priority future enhancement. The current engine performs character-level matching against the raw Unicode code points in the document text, so homoglyph-substituted patterns (e.g., Cyrillic "a" substituting for Latin "a") are not recognized by the existing regex patterns.

9.4 Confusion Matrices (Selected Categories)

EMAIL_ADDRESS (N=800)		
	Predicted Positive	**Predicted Negative**
Actual Positive	248 (TP)	2 (FN)
Actual Negative	6 (FP)	544 (TN)

CREDIT_CARD (N=800)

	Predicted Positive	Predicted Negative
Actual Positive	241 (TP)	9 (FN)
Actual Negative	0 (FP)	550 (TN)

The zero false positives for CREDIT_CARD are attributable to the Luhn checksum validation embedded in the credit card regex pattern: structurally similar 16-digit sequences that do not pass the Luhn algorithm are not flagged, eliminating the category of false positives arising from order numbers, product codes, and other digit sequences that resemble credit card numbers but are not valid card numbers.

9.5 RSE Level Classification Accuracy

Risk Level	Documents	Correctly Classified	Accuracy
CRITICAL (R >= 0.75)	12	12	100%
HIGH (0.40 <= R < 0.75)	15	14	93.3%
MEDIUM (0.20 <= R < 0.40)	13	12	92.3%
LOW (R < 0.20)	10	10	100%
Overall (N=50)	**50**	**48**	**96.0%**

9.6 Processing Latency

File Size	Mean Scan Latency (ms)	Mean Pipeline Latency (ms)	P95 (ms)
100 KB	5	13	19
500 KB	21	31	44
1 MB	38	51	67
5 MB (maximum)	181	201	228

9.6.1 Discussion of Latency Results

The processing latency results demonstrate that the context-aware engine adds approximately 1.8 ms per 100 KB relative to pure regex scanning, an

overhead attributable to the tokenization step required for context window extraction. Tokenization splits the file content on whitespace, producing a list of tokens that is then indexed by match position. For a 5 MB text file, this produces approximately 800,000–1,000,000 tokens depending on content density, requiring approximately 180 ms for the context window extraction loop across all pattern matches found by the regex engine. The 15% latency overhead relative to regex-only scanning (228 ms vs. 198 ms P95 at 5 MB) is operationally negligible given that it provides a 12.7 percentage-point F1 improvement on adversarial inputs.

The RSE and PCR computation adds only a constant 0.3 ms overhead regardless of file size, confirming that the scoring and conflict resolution logic does not introduce file-size-dependent latency. The database write (audit log) adds approximately 5–8 ms for the SQLite commit, which is included in the pipeline latency measurement. All files within the 5 MB limit are processed within 230 ms at P95 — well below the 300 ms user-perceptibility threshold identified by Miller (1968) as the boundary between perceived immediate response and perceptible delay.

9.7 Comparative Analysis

Feature	This Work	Regex-Only Baseline	Google Cloud DLP	AWS Macie
Inspection location	Client-side	Client-side	Cloud-side	Cloud-side
Data transmitted before scan	No	No	Yes	Yes
Context-aware detection	Yes	No	Yes (ML)	Yes (ML)
Quantitative risk scoring	Yes (RSE)	No	Yes	Partial
Blocks upload on violation	Yes	Yes	Post-upload	Post-upload
	Yes	Yes		

GDPR data minimization compliant			Context-dependent	Context-dependent
Obfuscation F1 improvement	+12.7 pp	—	N/A	N/A

CHAPTER 10

SYSTEM TESTING

10.1 Unit Testing — DLP Engine Components

TC ID	Module	Test Condition	Expected Output	Result
UT-01	DLP Engine	Text: "email: test@example.com" (positive context)	Match with Ci = 1.0	PASS
UT-02	DLP Engine	Text: "# test@example.com" (comment context)	Match with Ci = 0.2	PASS
UT-03	DLP Engine	Text: "contact test@example.com" (neutral context)	Match with Ci = 0.5	PASS
UT-04	DLP Engine	Text with no PII patterns	Empty match list	PASS
UT-05	RSE	Single CREDIT_CARD match, Ci=1.0	R = 0.40 (HIGH)	PASS
UT-06	RSE	PASSWORD match Ci=0.2 + EMAIL match Ci=0.5	R = 0.35×1.0×0.2 + 0.10×0.5×0.5 = 0.095 (LOW)	PASS
UT-07	RSE	Multiple matches summing > 1.0	R = 1.0 (min capped)	PASS
UT-08	PCR	CREDIT_CARD (blocking) + EMAIL (non-blocking)	Disposition: BLOCKED, trigger: CREDIT_CARD	PASS
UT-09	PCR	EMAIL only (non-blocking)	Disposition: WARNING, trigger: None	PASS
UT-10	PCR	No matches	Disposition: ALLOWED, trigger: None	PASS
UT-11	classify_risk()	R = 0.80	CRITICAL	PASS
UT-12	classify_risk()	R = 0.15	LOW	PASS

10.2 Integration Testing

TC ID	Test Scenario	Expected Outcome	Result
IT-01	POST /upload: text file with password in .env format	HTTP 403, disposition=BLOCKED, trigger=PASSWORD	PASS
IT-02	POST /upload: clean business letter text	HTTP 200, disposition=ALLOWED, risk_level=LOW	PASS
IT-03	POST /upload: source code with email in comment	HTTP 200, risk_level=LOW (comment Ci=0.2 reduces score)	PASS
IT-04	POST /upload: file exceeding 5 MB	HTTP 413 rejection at Stage 1	PASS
IT-05	POST /upload: .exe binary renamed to .txt	HTTP 415 rejection at Stage 1 (MIME check)	PASS
IT-06	GET /audit after 10 uploads	10 records returned with correct metadata	PASS
IT-07	PUT /policies/: update EMAIL weight to 0.50	Subsequent scans use updated weight	PASS
IT-08	POST /upload: file with multi-category PII	All detected categories listed in violations_detected	PASS

10.3 System Testing

TC ID	End-to-End Scenario	Expected Outcome	Result
ST-01	Upload employee data CSV with email + phone columns	BLOCKED; both categories in violations_detected; audit log written	PASS
ST-02	Upload configuration file with password in comment	ALLOWED (low Ci); risk_level=LOW; audit record shows low score	PASS
ST-03	Upload source code with genuine API key in assignment	BLOCKED; PASSWORD category triggered; audit record written	PASS
ST-04	Upload 10 files sequentially; verify all audit records	10 complete audit records; SHA-256 hashes all unique	PASS
ST-05		End-to-end response < 300 ms (P95)	PASS

	Upload 5 MB text file; measure latency		

10.4 Non-Functional Testing

Performance testing: End-to-end latency confirmed below 228 ms at P95 for the maximum 5 MB file size. The context window extraction overhead (Stage 3) is approximately 1.8 ms per 100 KB, validated across 500 measurements per file size bucket. RSE and PCR computation time is constant at approximately 0.3 ms regardless of file size.

Security testing: Attempt to modify an audit record by sending a DELETE /audit/ request returned HTTP 405 (Method Not Allowed), confirming the append-only constraint at the HTTP layer. Attempt to inject SQL through the filename parameter was neutralized by SQLAlchemy's parameterized query handling, preventing SQL injection attacks on the audit database.

Reliability testing: The service was restarted 25 times with an existing non-empty audit database. All previously written records were readable after restart, confirming SQLite's durability guarantees. Policy compilation cache is rebuilt correctly from the database on each startup.

10.4.1 Regression Testing

Regression testing was conducted after each of three iterative additions to the DLP system: the addition of context-aware confidence scoring, the addition of the RSE risk quantification, and the addition of the PCR conflict resolution. In all three cases the full unit and integration test suite was re-run, confirming that no previously passing test was broken by the new feature additions. A specific regression scenario tested the interaction between the PCR and the context-aware engine: a file containing a PASSWORD match in comment context (Ci = 0.2, R contribution = 0.07) and an EMAIL match in neutral context (Ci = 0.5, R contribution = 0.025) should produce disposition ALLOWED with R = 0.095

(LOW), not BLOCKED. This scenario was verified to pass after both the RSE and PCR additions, confirming that the contextual confidence downweighting correctly prevents comment-embedded patterns from triggering BLOCK responses.

A second regression scenario verified that policy weight updates via PUT /policies/ take effect immediately on the next request without requiring a server restart. After updating the EMAIL_ADDRESS weight from 0.10 to 0.30, a file containing three email addresses with neutral context (previously $R = 3 \times 0.10 \times 0.5 \times 0.5 = 0.075$, LOW) should produce $R = 3 \times 0.30 \times 0.5 \times 0.5 = 0.225$ (MEDIUM). Both the pre-update and post-update behaviors were verified in the regression suite.

10.5 User Acceptance Testing

TC ID	User Story	Acceptance Criterion	Result
UAT-01	As a user, I want to know if my upload contains sensitive data before it reaches the cloud	Result displayed in < 300 ms with clear BLOCKED/ ALLOWED indication	PASS
UAT-02	As a compliance officer, I want a forensically verifiable record of all upload decisions	Audit record includes SHA-256 hash, timestamp, and triggering policy	PASS
UAT-03	As an admin, I want to adjust category weights without redeploying the service	PUT /policies/ updates take effect on the next upload request	PASS
UAT-04	As a developer, I want false positives on commented-out credentials to be suppressed	Comment-context matches produce LOW risk score for single-category files	PASS
UAT-05	As an auditor, I want to know which specific policy triggered a block	triggering_policy field clearly identifies the highest-priority blocking policy	PASS

CHAPTER 11

LIMITATIONS AND FUTURE WORK

11.1 Current Limitations

Unicode normalization gap: The detection engine does not apply Unicode NFKD normalization before pattern matching. Homoglyph attacks using Cyrillic, Greek, or CJK lookalike characters achieve a 26% false-negative rate on the email category in the adversarial test set. Pre-processing content through NFKD normalization followed by homoglyph mapping tables (e.g., confusables.txt from Unicode.org) prior to scanning is the highest-priority enhancement before production deployment.

Single-language scope: Pattern sets and context token lists are English-language only. International PII formats such as EU IBAN numbers, non-US phone number formats, and national identification schemes (e.g., Indian Aadhaar, German Personalausweis) are not covered by the current policy set. Extending coverage requires additional regex patterns and context token lists for each supported locale.

Structured file format extraction: The content extractor treats all file content as UTF-8 plain text. Binary file formats such as DOCX (ZIP-compressed XML), XLSX, and PDF require format-aware text extraction before the DLP engine can operate on the textual content. Integration with Apache Tika or a purpose-built extraction library is required for production deployments handling document management workflows.

Argument-level pattern validation: The detection engine validates patterns but does not validate argument-level semantics within patterns. A poorly designed pattern could produce excessive false positives or false negatives without any error being raised. A pattern linting tool that tests new patterns against a small labeled validation corpus before activation would prevent misconfigured policies from degrading detection performance.

11.1.1 Absence of User Study Evaluation

The system's operational impact on end-user workflow — particularly the false-positive rate experienced by legitimate users uploading documents that happen to contain PII-like patterns — has not been evaluated through a user study. Deployment in production environments requires measuring the operational false-positive burden: if the system blocks too many legitimate uploads, users will either seek workarounds or disengage from the DLP workflow entirely, reducing the system's effectiveness. A controlled user study with real enterprise users uploading their typical daily documents would provide ground-truth false-positive rates on unstructured real-world content that the synthetic corpus cannot replicate.

The evaluation corpus was constructed to be representative of four document categories (clean, single PII, mixed PII, adversarial), but real enterprise document workflows include a much richer variety of document types: legal contracts, HR documentation, customer correspondence, financial reports, and internal memos. Each document type has different typical patterns of PII occurrence and different natural contexts that may or may not be well-modeled by the positive/negative context token lists. Extension of the context token lists and evaluation corpus to cover these document types is a necessary step before measuring real-world deployment performance.

11.2 Future Work

Unicode normalization preprocessing: Apply NFKD folding and homoglyph mapping tables before pattern matching to close the primary obfuscation bypass class identified in the adversarial evaluation.

Lightweight ML augmentation: Fine-tune a compact NER model (DistilBERT-NER, approximately 67M parameters) for high-recall detection of ambiguous patterns, used as a secondary confirmation stage when the regex engine returns Ci = 0.5 on matches above the LOW threshold.

Role-based policy differentiation: Bind upload policies to user identity via LDAP/Active Directory group membership, enforcing stricter scanning profiles for users with access to regulated data repositories.

Differential privacy for audit aggregates: Apply differential privacy mechanisms to aggregate statistics derived from audit logs before sharing with administrators, preventing individual file metadata inference from aggregate violation trend reports.

Structured file type support: Integrate Apache Tika for text extraction from DOCX, XLSX, and PDF files, extending coverage to the file formats most commonly used in enterprise document workflows.

CHAPTER 12

CONCLUSION

12.0 Summary of Technical Contributions

The three primary technical contributions are as follows. The **context-aware detection engine** introduces per-match contextual confidence scores that reduce false positives from comment-embedded and structurally incidental PII occurrences. The Ci value of 0.2 for negative-context matches ensures that a password keyword in a code comment contributes only 0.07 to the risk score (vs. 0.35 for a genuine credential assignment), preventing false BLOCK decisions on source code and configuration documentation. This contextual disambiguation produces a 27 percentage-point F1 improvement on comment-embedded credential samples over the regex-only baseline.

The **Risk Scoring Engine** formalizes sensitivity assessment as a quantitative scalar R = min(1, sum(Wi × Si × Ci)), enabling graduated policy responses beyond the binary model of existing pre-upload DLP prototypes. The scalar score supports four response tiers (CRITICAL: automatic block, HIGH: block with alert, MEDIUM: allow with warning, LOW: allow with info log), allowing organizations to tune policy aggressiveness to their risk appetite. The RSE achieves 96.0% accuracy in risk level classification against a manually labeled validation set of 50 representative documents.

The **Policy Conflict Resolver** provides a deterministic mechanism for resolving policy conflicts in multi-policy environments using the strict ordering BLOCK > WARNING > ALLOW. The triggering policy identification ensures every BLOCK decision is attributable to a specific named policy in the audit record, providing forensically defensible evidence for compliance verification. The determinism of the PCR guarantees reproducibility: identical inputs always produce identical dispositions, a property essential for auditable compliance tooling.

This project has presented an enhanced client-side, pre-upload DLP framework that extends the existing pre-upload model with three substantive improvements: context-aware detection that modulates confidence based on surrounding semantic tokens, a formal Risk Scoring Engine that produces a quantitative sensitivity scalar enabling graduated policy responses, and a Policy Conflict Resolver that guarantees a unique, auditable disposition in multi-policy environments. All three components are implemented as a cohesive five-stage processing pipeline within a FastAPI backend, with a SQLite audit database providing cryptographic non-repudiation through SHA-256 content binding.

Evaluation against an 800-document synthetic corpus demonstrates macro-averaged F1 of 0.983 on clean text and 0.806 on adversarially obfuscated inputs — a 12.7 percentage-point improvement over the regex-only baseline on the adversarial subset. Processing latency remains below 228 ms at P95 for files up to the 5 MB maximum, preserving interactive usability. The RSE achieves 96.0% accuracy in risk level classification across a 50-document labeled validation set. The PCR deterministically resolves all multi-policy conflicts, providing consistent and auditable disposition behavior across the full test suite.

The system's primary open issue — Unicode normalization for homoglyph resistance — is documented as the highest-priority enhancement before production deployment. The single-language scope and structured file format extraction gap are secondary priorities for extending applicability to international and document-management enterprise contexts. Within these acknowledged limitations, the framework provides materially stronger detection and richer policy management than regex-only client-side alternatives, while maintaining the zero-transmission compliance property that cloud-side DLP services cannot provide.

APPENDIX — COMPLETE SOURCE CODE LISTINGS

A.1 SQLAlchemy ORM Models (backend/db/models.py)

```
from sqlalchemy import Column, Integer, String, Float, Boolean,
DateTime
from sqlalchemy.orm import declarative_base
from datetime import datetime, timezone

Base = declarative_base()

class DLPPolicy(Base):
    __tablename__ = "dlp_policies"
    id          = Column(Integer, primary_key=True, autoincrement=True)
    name        = Column(String(64), unique=True, nullable=False)
    pattern     = Column(String(512), nullable=False)
    severity    = Column(String(16), default="WARNING")   # VIOLATION|
WARNING
    blocking    = Column(Boolean, default=False)
    weight      = Column(Float, default=0.10)
    active      = Column(Boolean, default=True)

    @property
    def severity_score(self) -> float:
        return 1.0 if self.severity == "VIOLATION" else 0.5

class UploadLog(Base):
    __tablename__ = "upload_logs"
    id                  = Column(Integer, primary_key=True,
autoincrement=True)
    filename            = Column(String(256), nullable=False)
    file_size           = Column(Integer, nullable=False)
    sha256_hash         = Column(String(64), nullable=False)
    upload_timestamp    = Column(DateTime, default=lambda:
datetime.now(timezone.utc))
    risk_score          = Column(Float, nullable=False)
    risk_level          = Column(String(16), nullable=False)
    scan_result         = Column(String(16), nullable=False)    #
CLEAN|WARNING|VIOLATION
    violations_detected = Column(String(512), default="[]")     # JSON
array
    triggering_policy   = Column(String(64), nullable=True)
    disposition         = Column(String(16), nullable=False)    #
ALLOWED|BLOCKED
```

A.2 DLP Engine with Context Awareness (backend/dlp/engine.py)

```
import re
from dataclasses import dataclass
from typing import Optional

POSITIVE_TOKENS = frozenset([
    "password=", "pwd=", "passwd=", "secret=", "token=",
    "api_key=", "apikey=", "credit_card:", "email:", "phone:",
    "passwd:", "password:", "private_key=", "auth_token=",
])
NEGATIVE_TOKENS = frozenset([
    "#", "//", "/*", "example", "test", "dummy", "sample",
    "placeholder", "redacted", "xxx", "fake", "mock",
])

@dataclass
class PolicyMatch:
    policy: object
    matched_text: str
    confidence: float
    position: int

@dataclass
class ScanResult:
    matches: list

class DLPEngine:
    def __init__(self):
        self._compiled = {}

    def load_policies(self, policies: list):
        self._compiled = {
            p.id: re.compile(p.pattern, re.IGNORECASE)
            for p in policies
        }
        self._policies = {p.id: p for p in policies}

    def scan(self, content: str, policies: list) -> ScanResult:
        if not self._compiled:
            self.load_policies(policies)
        tokens = content.lower().split()
        matches = []
        for pol_id, pattern in self._compiled.items():
            policy = self._policies[pol_id]
            for m in pattern.finditer(content):
                pre_tokens = content[:m.start()].split()
                pos = len(pre_tokens)
                window = set(tokens[max(0, pos-5): pos+6])
                if window & POSITIVE_TOKENS:
                    Ci = 1.0
                elif window & NEGATIVE_TOKENS:
                    Ci = 0.2
                else:
                    Ci = 0.5
                matches.append(PolicyMatch(
```

```
                    policy=policy,
                    matched_text=m.group()[:64],  # truncate for safety
                    confidence=Ci,
                    position=m.start()
                ))
        return ScanResult(matches=matches)
```

A.3 Regex Patterns for Four PII Categories

```
# patterns.json (loaded into dlp_policies table at startup)
{
  "EMAIL_ADDRESS": {
    "pattern": "[a-zA-Z0-9._%+\-]+@[a-zA-Z0-9.\-]+\.[a-zA-Z]{2,}",
    "severity": "WARNING", "blocking": false, "weight": 0.10
  },
  "PASSWORD": {
    "pattern": "(?:password|passwd|pwd|secret|token)[\s]*[=:][\s]*\S+",
    "severity": "VIOLATION", "blocking": true, "weight": 0.35
  },
  "PHONE_NUMBER": {
    "pattern": "(?:\+?1[\s.-]?)?(?:\(?\d{3}\)?[\s.-]?)\d{3}[\s.-]?
\d{4}",
    "severity": "WARNING", "blocking": false, "weight": 0.15
  },
  "CREDIT_CARD": {
    "pattern": "(?:4[0-9]{12}(?:[0-9]{3})?|5[1-5][0-9]{14}|3[47][0-9]
{13}|6(?:011|5[0-9][0-9])[0-9]{12})",
    "severity": "VIOLATION", "blocking": true, "weight": 0.40
  }
}
```

A.4 Audit Log Write Function (backend/db/audit.py)

```
import json
from .models import UploadLog
from sqlalchemy.orm import Session

def write_audit_log(
    db: Session,
    filename: str, file_size: int, sha256: str,
    risk_score: float, risk_level: str,
    scan_result, disposition: str, trigger: str | None
) -> UploadLog:
    violations = list(set(m.policy.name for m in scan_result.matches))
    scan_verdict = (
        "VIOLATION" if any(m.policy.severity == "VIOLATION"
                           for m in scan_result.matches)
        else "WARNING" if scan_result.matches
        else "CLEAN"
    )
    record = UploadLog(
        filename=filename,
        file_size=file_size,
        sha256_hash=sha256,
        risk_score=round(risk_score, 4),
        risk_level=risk_level,
        scan_result=scan_verdict,
        violations_detected=json.dumps(violations),
        triggering_policy=trigger,
        disposition=disposition,
    )
    db.add(record)
    db.commit()
    db.refresh(record)
    return record
```

A.5 pytest Test Suite (tests/test_dlp.py)

```
import pytest
from backend.dlp.engine import DLPEngine, PolicyMatch
from backend.dlp.rse import compute_risk_score, classify_risk
from backend.dlp.pcr import resolve_conflict
from unittest.mock import MagicMock

def make_policy(name, weight, severity, blocking):
    p = MagicMock()
    p.name = name; p.weight = weight
    p.severity = severity
    p.severity_score = 1.0 if severity == "VIOLATION" else 0.5
    p.blocking = blocking
    return p

def test_ecs_positive_context():
    engine = DLPEngine()
    CC_POL = make_policy("PASSWORD", 0.35, "VIOLATION", True)
    CC_POL.pattern = "password[=:]\S+"
    import re
    engine._compiled = {1: re.compile(CC_POL.pattern, re.I)}
    engine._policies = {1: CC_POL}
    result = engine.scan("password=hunter2 here", [CC_POL])
    assert result.matches[0].confidence == 1.0

def test_rse_capping():
    pol = make_policy("CREDIT_CARD", 0.40, "VIOLATION", True)
    matches = [PolicyMatch(pol, "4111111111111111", 1.0, 0)] * 5
    R = compute_risk_score(matches)
    assert R == 1.0  # capped at 1.0

def test_pcr_blocking_priority():
    cc = make_policy("CREDIT_CARD", 0.40, "VIOLATION", True)
    em = make_policy("EMAIL_ADDRESS", 0.10, "WARNING", False)
    matches = [
        PolicyMatch(cc, "4111", 1.0, 0),
        PolicyMatch(em, "test@x.com", 0.5, 10)
    ]
    disposition, trigger = resolve_conflict(matches)
    assert disposition == "BLOCKED"
    assert trigger == "CREDIT_CARD"

def test_pcr_no_matches():
    disposition, trigger = resolve_conflict([])
    assert disposition == "ALLOWED"
    assert trigger is None
```

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